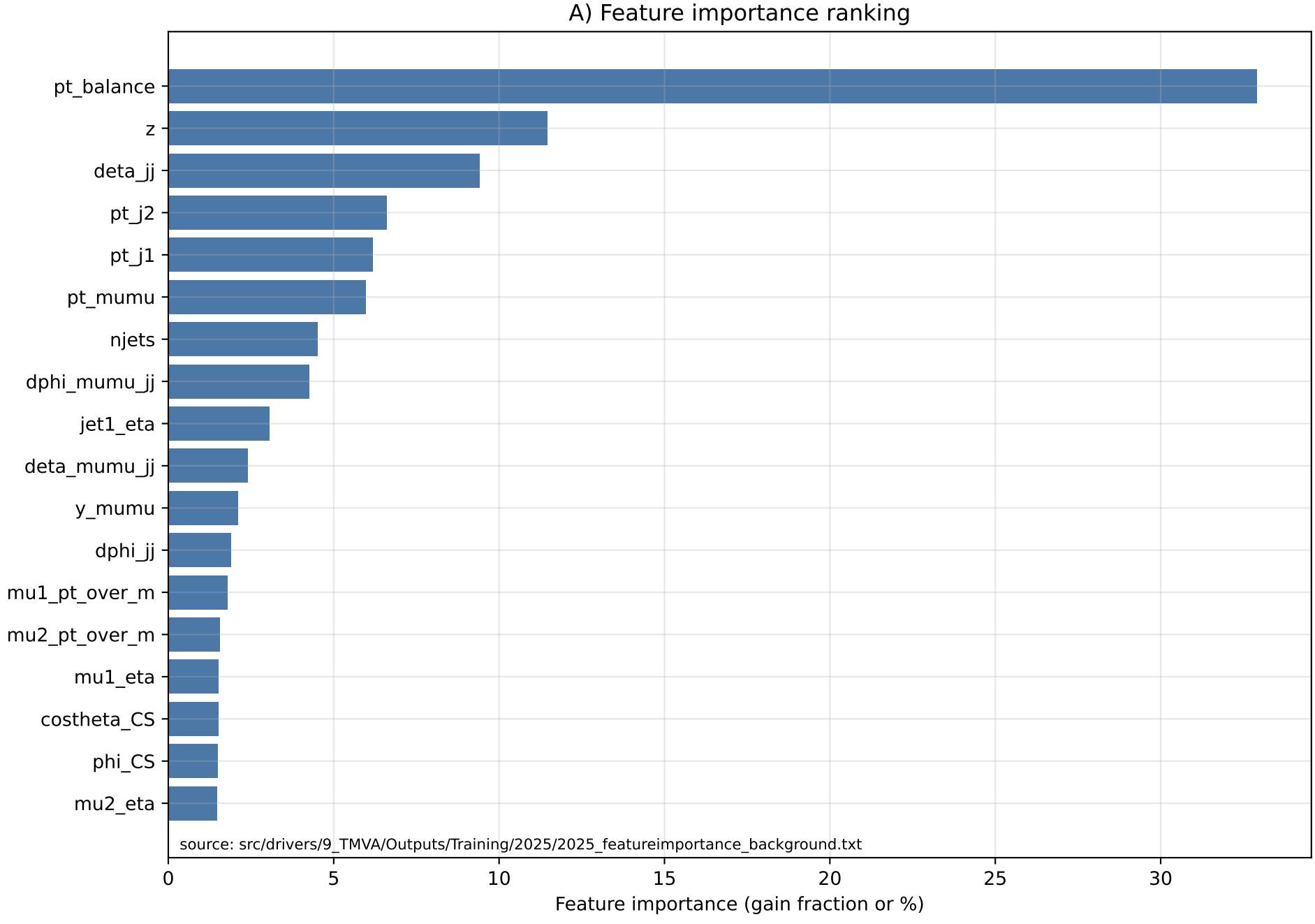
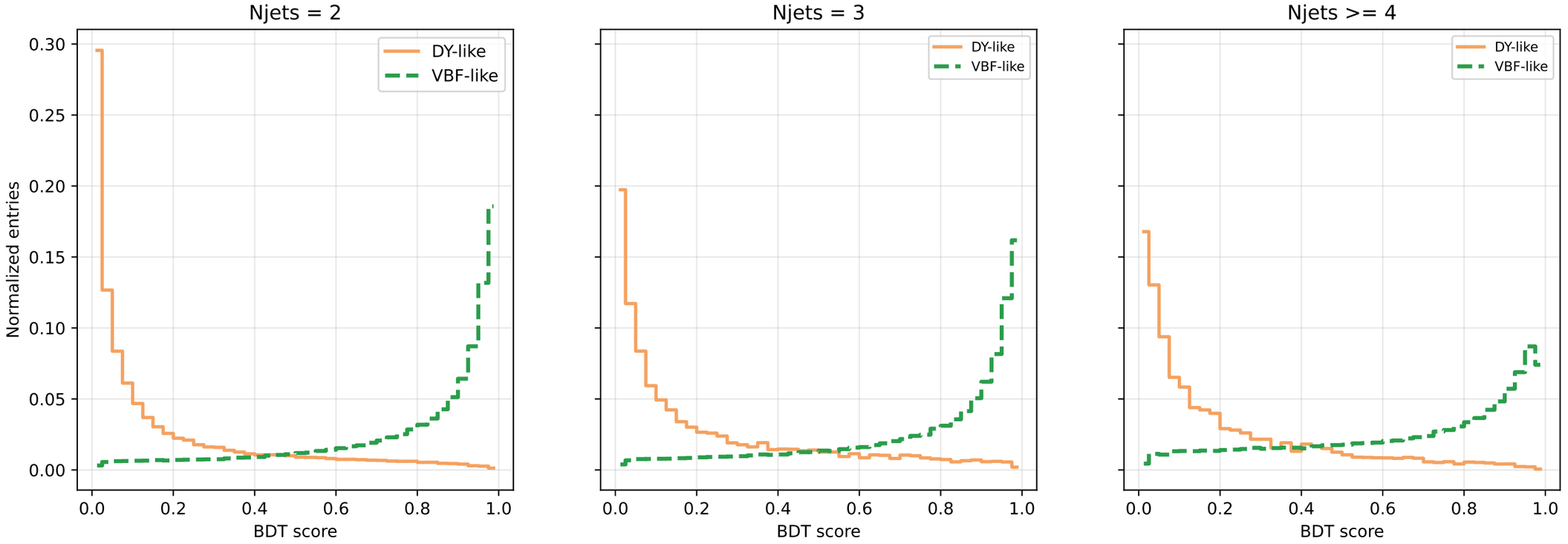
VBF BDT Validation - 2025

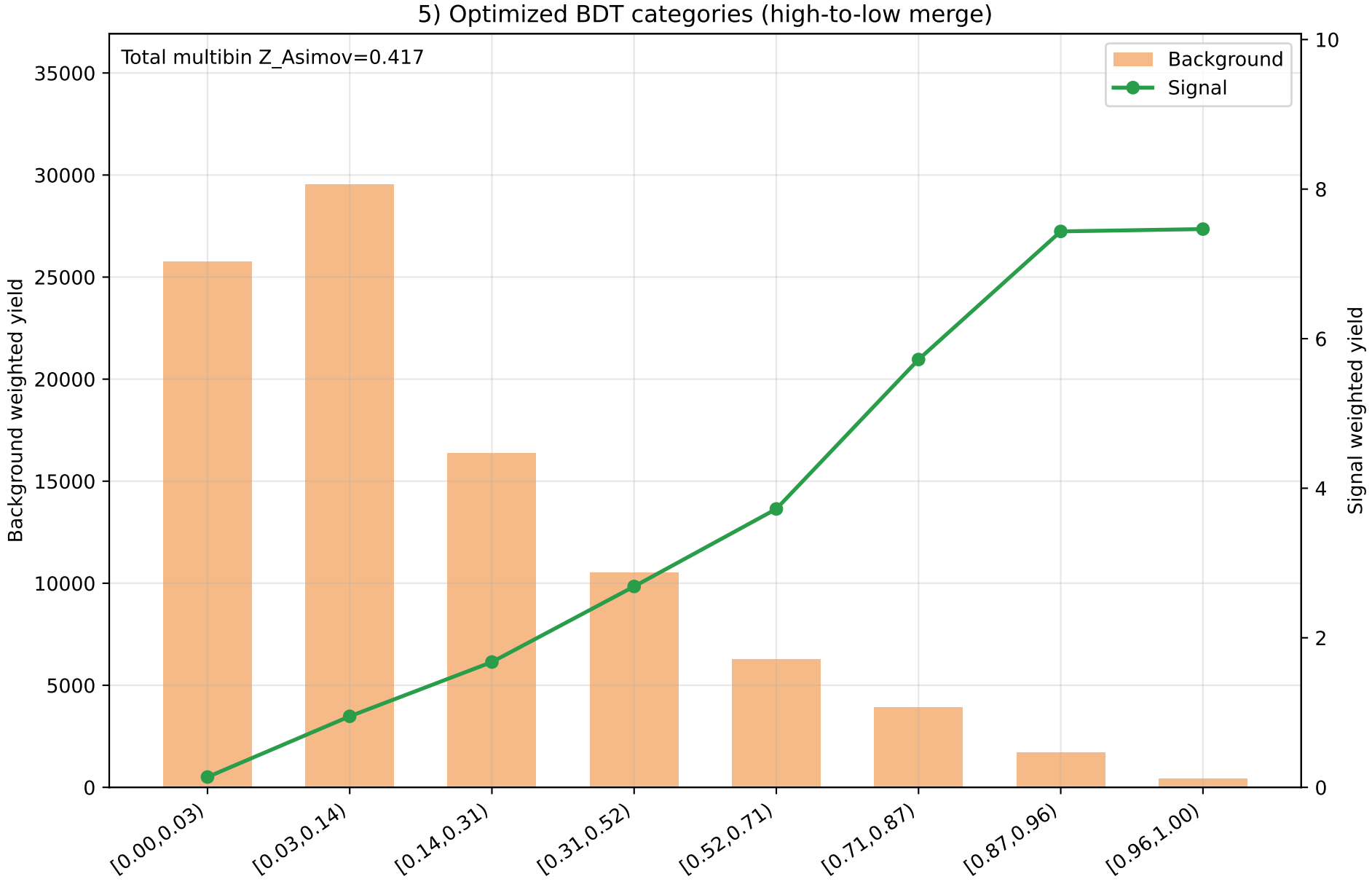
```
Standalone per-era diagnostic script: bdt_validation.py
Logic references: ScanYields / PlotScored / MassSculpt / KS_perfold / DataMC
Resolved branches: {'score': 'bdt_score', 'weight': 'weight', 'mass': 'm_mumu', 'njets': 'njets', 'mjj': None, 'de
```

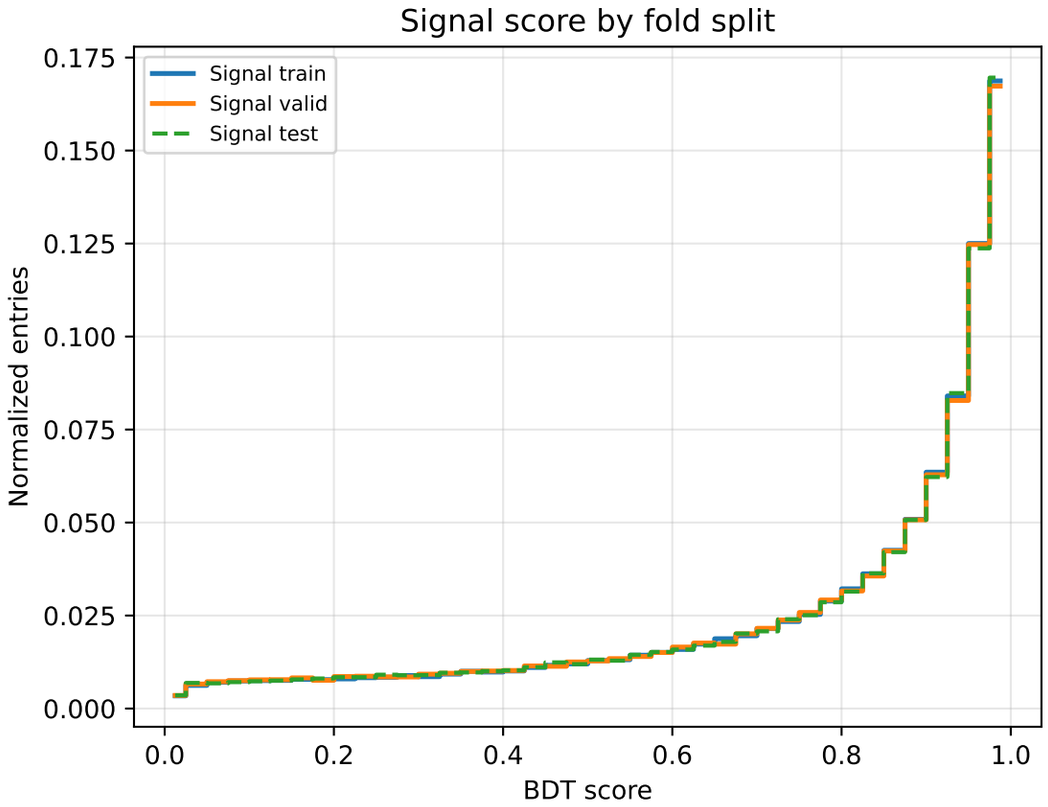
Requested VBF feature cross-check

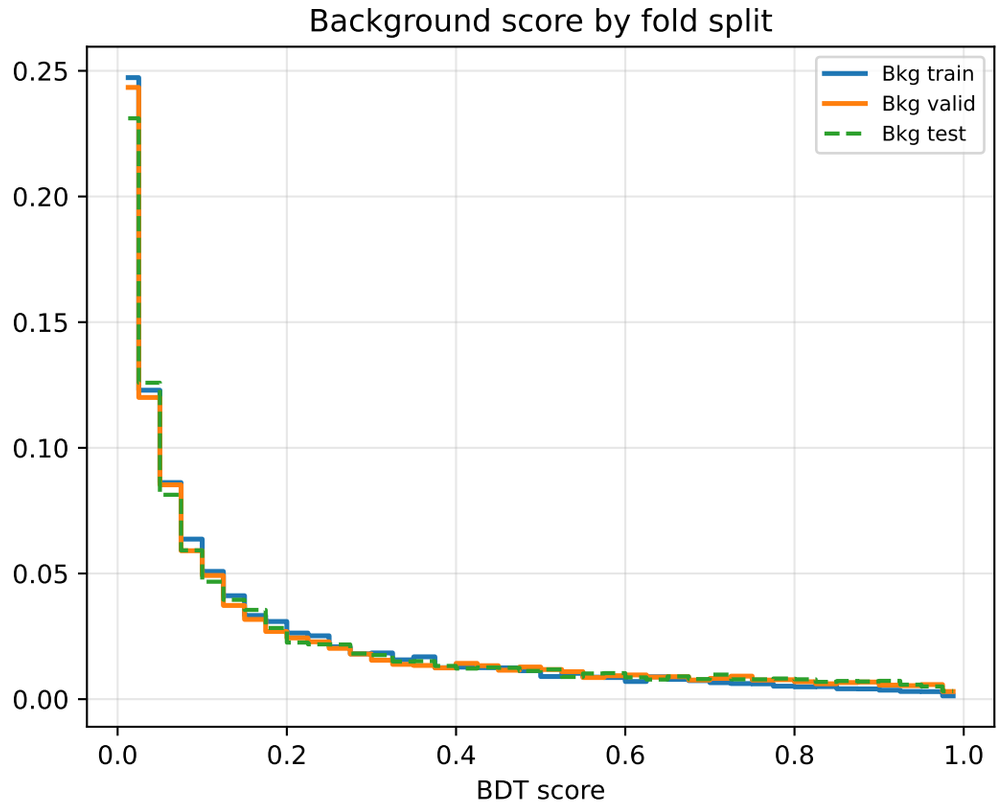
requested	status	resolved_as	notes
jet1_qgl	missing	-	not found in tree branches
jet1_eta	exact	jet1_eta	
jet2_eta	proxy	y_j2	rapidity, not eta
jet1_pt	alias	pt_j1	
jet2_pt	alias	pt_j2	
mjj	missing	-	not found in tree branches
dEta_jj	alias	deta_jj	
dPhi_jj	alias	dphi_jj	
mmj_min_dEta	proxy	deta_mumu_jj	dimuon-dijet deta, not min(mu,jet)
mmj_min_dPhi	proxy	dphi_mumu_jj	dimuon-dijet dphi, not min(mu,jet)
pt_centrality	proxy	z	zeppenfeld variable, not pt centrality
dimuon_pt	alias	pt_mumu	
dimuon_rapidity	alias	y_mumu	
dimuon_cos_theta_cs	alias	costheta_CS	
dimuon_phi_cs	alias	phi_CS	
dimuon_mass_resolution	missing	-	not found in tree branches
mu1_pt_over_mass	alias	mu1_pt_over_m	
mu2_pt_over_mass	alias	mu2_pt_over_m	
mu1_eta	exact	mu1_eta	
mu2_eta	exact	mu2_eta	
N_jets	alias	njets	
year	derived	year=2025	from --era


A) Feature importance ranking
pt_balance
z
deta_jj
pt_j2
pt_j1
pt_mumu
njets
dphi_mumu_jj
jet1_eta
deta_mumu_jj
y_mumu
dphi_jj
mu1_pt_over_m
mu2_pt_over_m
mu1_eta
costheta_CS
phi_CS
mu2_eta
source: src/drivers/9_TMVA/Outputs/Training/2025/2025_featureimportance_background.txt
0
5
10
15
20
25
30
Feature importance (gain fraction or %)


BDT score vs jet multiplicity (SR)
Njets = 2
Njets = 3
Njets >= 4
DY-like
VBF-like
Normalized entries
BDT score
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.0
0.2
0.4
0.6
0.8
1.0


5) Optimized BDT categories (high-to-low merge)
Total multibin Z_Asimov=0.417
Background
Signal
Background weighted yield
Signal weighted yield
0
5000
10000
15000
20000
25000
30000
35000
0
2
4
6
8
10
[0.00,0.03)
[0.03,0.14)
[0.14,0.31)
[0.31,0.52)
[0.52,0.71)
[0.71,0.87)
[0.87,0.96)
[0.96,1.00)



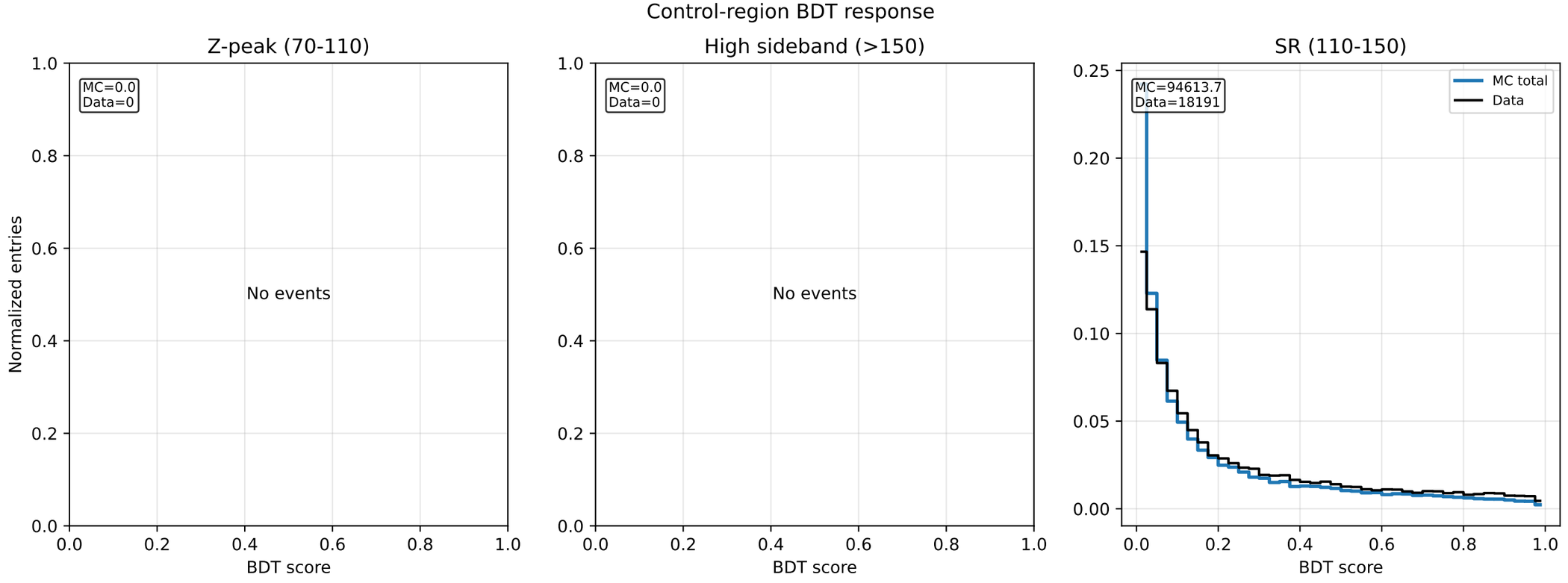


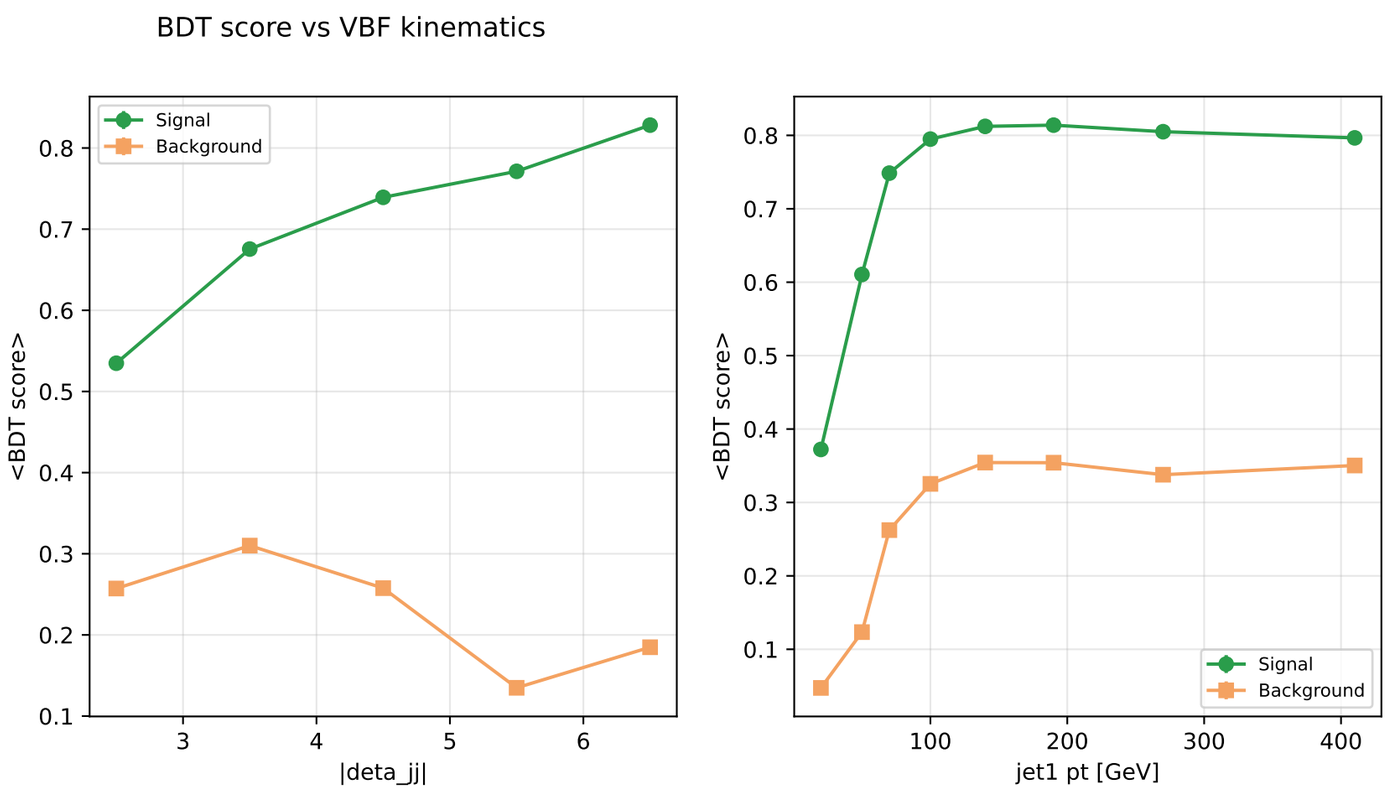
Fold-based KS checks (train vs validation)

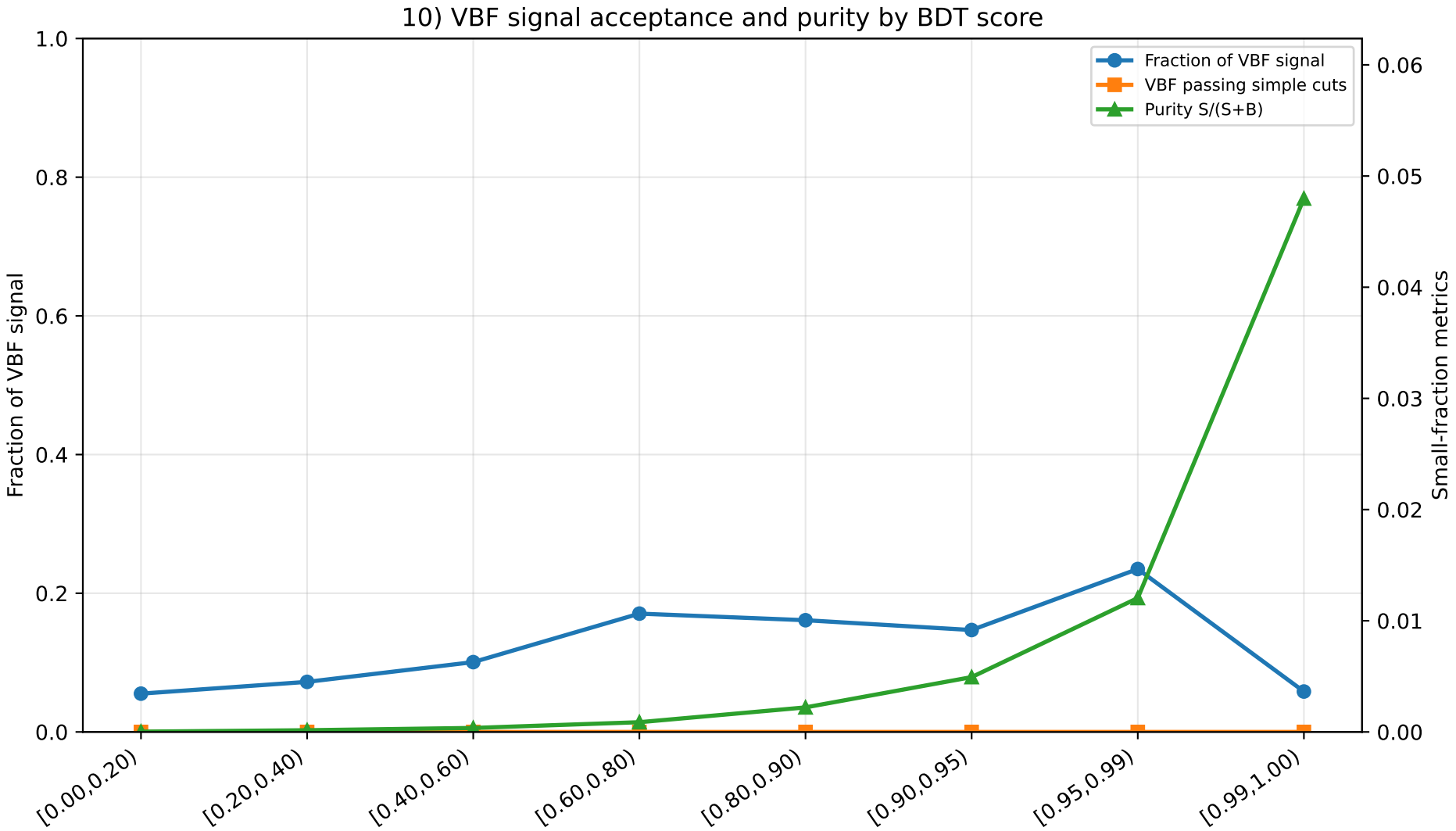
KS thresholds: Run2-loose: p>=0.01, D<=0.15; Run2-nominal: p>=0.05, D<=0.1; Run2-tight: p>=0.1, D<=0.07

Sample	N train	N valid	N test	Mean(train)	Mean(valid)	Mean(test)	KS D	KS p	Run2-loose	Run2-nominal	Run2-tight
Signal	800334	400167	400166	0.7619	0.7591	0.7603	0.0044	5.461e-05	**FAIL**	**FAIL**	**FAIL**
Background	218920	109454	109454	0.1907	0.2125	0.2171	0.0114	1.285e-08	**FAIL**	**FAIL**	**FAIL**

Corr(score, njets): signal=-0.130, background=0.052


Control-region BDT response
Z-peak (70-110)
High sideband (>150)
SR (110-150)
MC=0.0
Data=0
No events
MC=0.0
Data=0
No events
MC=94613.7
Data=18191
MC total
Data
Normalized entries
BDT score
BDT score
BDT score
0.0
0.2
0.4
0.6
0.8
1.0
0.00
0.05
0.10
0.15
0.20
0.25


BDT score vs VBF kinematics
Signal
Background
<BDT score>
|deta_jj|
jet1 pt [GeV]


10) VBF signal acceptance and purity by BDT score
Fraction of VBF signal
VBF passing simple cuts
Purity S/(S+B)
Fraction of VBF signal
Small-fraction metrics
1.0
0.8
0.6
0.4
0.2
0.0
0.06
0.05
0.04
0.03
0.02
0.01
0.00
[0.00,0.20)
[0.20,0.40)
[0.40,0.60)
[0.60,0.80)
[0.80,0.90)
[0.90,0.95)
[0.95,0.99)
[0.99,1.00)

Validation Summary

Era: 2025
Inputs: data_files=1 mc_groups=7
Signal groups: SIGNAL
VBF groups: SIGNAL
SR weighted yields: S=29.794, B=94583.864
KS profiles: Run2-loose: p>=0.01, D<=0.15; Run2-nominal: p>=0.05, D<=0.1; Run2-tight: p>=0.1, D<=0.07
Feature importance source: src/drivers/9_TMVA/Outputs/Training/2025/2025_featureimportance_background.txt
Optimized categories: n=8, Z_Asimov=0.4168
KS(signal train-vs-valid): D=0.0044, p=5.461e-05
KS(bkg train-vs-valid): D=0.0114, p=1.285e-08
KS profile checks (Signal): Run2-loose=FAIL, Run2-nominal=FAIL, Run2-tight=FAIL
KS profile checks (Background): Run2-loose=FAIL, Run2-nominal=FAIL, Run2-tight=FAIL
Corr(score, njets) signal=-0.130, background=0.052
Highest BDT-bin purity=0.048